

Leveraging Large Language Models for Usability Testing: a Preliminary Study

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Abstract

Despite growing efforts to prioritize user experience in product development, software organizations often perform little or no usability engineering activities. Therefore, it is crucial to develop strategies to integrate them effectively into software development processes. The rapid advances in Artificial Intelligence have significantly influenced various aspects of daily life, particularly with the emergence of Large Language Models (LLMs), which can serve as promising tools to support activities to enhance the usability of software products. This paper presents a study investigating the potential of LLMs to assist practitioners in conducting usability tests. Specifically, we conducted an experiment where LLMs generate usability test tasks. Our goal is to assess whether AI can effectively support evaluators by comparing tasks generated by LLMs to those defined by usability experts. The findings indicate that while LLMs can provide valuable support, effective usability testing still requires human oversight and expert intervention.

CCS Concepts

• **Human-centered computing** → **Usability testing**.

Keywords

Human-Centred Design, Large Language Models, Usability Test, Artificial Intelligence

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1 Introduction

The large and broad impact of Artificial Intelligence (AI) on modern society is reshaping how we carry out our daily activities in countless domains. Among the different types of AI, Large Language Models (LLMs) have received great attention from the general public in recent years, especially with the release of commercially-successful tools such as OpenAI's ChatGPT, Google's Gemini, and Microsoft's Copilot [7]. Making this technology more accessible has significantly changed the direction of research in academia and industry [3] since it can optimize and refine processes by exploiting its high computational power [9, 11]. Although Computer Science is often seen as the field that designs and creates new AI systems, it can also be on the receiving end. The extensive use of natural language in LLM-powered tools can facilitate and optimise the work of developers throughout the software development processes [3]. Researchers are investigating the impact of LLMs on the evaluation of systems' usability to understand to what extent experts can be supported [6, 13].

In this article, we present an experiment we conducted to explore whether and how LLMs can support usability experts during the design of the evaluation phase and, more specifically, in performing usability tests.

Despite the documented benefits of usability engineering, there are studies in the literature that show that still too many software development organisations either neglect usability and UX or do not consider them properly [1, 10, 12]. Among the reasons given by practitioners for the problems they encountered in introducing and performing usability evaluation in their software development processes, the most reported are the lack of appropriate usability evaluation methodologies and the fact that they require a lot of resources in terms of cost, time and people involved [1]. Based on these considerations, we are investigating if LLMs can serve as promising tools to support evaluation activities to enhance the usability of software products. Thus, the following Research Question (RQ) was formulated: *Can LLM-based tools support practitioners in designing usability tests?* We experimented with three LLM-based

tools (i.e. ChatGPT, Mistral, Gemini) prompting them to generate the tasks for a user study.

2 Methodology

Three LLM-based tools, i.e., *ChatGPT*, *Mistral* and *Gemini*, were prompted to generate a set of tasks to be administered to a sample of users. These tasks were then compared with those defined by a team of usability experts used in a previous study presented in [4]. The details of the two studies are described below.

2.1 Human-Led Evaluation Study

The study was conducted in the challenging context of medical cytology, addressing the problem of AI-assisted microscopic analysis of cells in the nasal mucosa. In particular, an AI-based system called Rhino-Cyt was evaluated to assess its ability to assist rhinocytologists in making nasal cytological diagnoses. Rhino-Cyt exploits a deep neural network that automatically classifies cells in a nasal cytology preparation. In addition, the system allows physicians to validate or correct the cell classification by manually intervening in case of errors or inaccuracies. It also includes explainability techniques that allow end-users to understand the motivations behind the results and facilitate the validation process based on their expertise [4]. Ensuring that domain experts are able to use this system correctly is crucial for effective collaboration between humans and AI, which means that both parties are capable to understand each other correctly through a symbiotic interaction paradigm that allows for their improvement and augmentation [5, 8]. The study involved a group of 9 doctors from different Italian hospitals who were asked to complete four tasks. These tasks were designed to observe user behaviour and preferences regarding the main system features, i.e. the clarification and intervention functions of the AI model. The tasks were:

- (T1) Check the classification of cells of the class “muciparous” and correct any errors.
- (T2) Check the classification of class “eosinophils” cells and correct any errors.
- (T3) Identify misclassified cells in class “neutrophils” and correct any errors.
- (T4) Identify misclassified cells in class “ciliated” and correct misclassifications and explanations, if any.

2.2 LLM-powered usability test generation

To evaluate the capability of LLMs to generate suitable usability testing tasks, we structured the experiment into three phases: 1. prompt creation; 2. task generation and analysis; and, 3. interview with an HCI expert. In the first phase, as previously mentioned, three platforms were used, and each model was configured with different parameters to test their randomness and limit potential hallucinations that could damage the outputs. The models and parameters were:

- (1) **ChatGPT:** Models *gpt-4o* and *gpt-4-turbo* were employed. *Temperature* was set to 0, 0.5, and 1 for each experiment.
- (2) **Mistral AI:** The models *mistral-medium-latest* and *mistral-large-latest* were tested with temperature values equal to 0, 0.5, and 1 (the same of ChatGPT) to maintain consistency in methodology across the different models.

- (3) **Gemini:** *gemini-1.5-pro* and *gemini-1.5-flash* were used with no temperature adjustments as these models did not support such configuration at the time of the experiment.

The same prompt was given as input for the three platforms with each configuration along with a description and screenshots of the Rhino-Cyt system to provide context and reduce potential mistakes. The prompt asks the LLM to generate five tasks that participants can carry out during a user study. Although the human-led study involved only four tasks, this experiment includes five tasks to comply with the best practices of HCI. The prompt was designed and redefined iteratively by a group of three experts in AI and HCI. Its final version is reported below:

You are an expert in usability testing for software systems. Your task is to generate tasks for a user test conducted with participants belonging to the category of rhinocytologists. The number of tasks must not exceed 5. Each task must meet the following criteria: provide clear instructions on what participants need to do; avoid using words that appear in the product interface; do not specify the steps required to complete the task, such as the name of the corresponding link or references to link text; be atomic, meaning each task must have a single, clear objective. To generate the tasks, follow these steps: analyze the description of the system to be tested, as indicated by the system description in the previous message; review the images of the interface included in the message; identify the system main functionalities, excluding typical features of systems in its category (e.g., registration or login functionality for web applications); create tasks that allow testing of the system primary functionalities; ensure that the generated tasks meet the specified criteria.

In the second phase, a group of the three usability experts analyzed the resulting tasks generated by the LLMs with the different configurations by taking into account their *Feasibility*, which is a binary value that indicates whether the task refers to a functionality that is actually provided by the system. Then, considering only the feasible tasks, the cosine similarity between the LLMs-generated and the human-generated tasks was computed and a subset of the 30 most similar ones (i.e. 5 tasks for each model) was selected for the next phase.

Finally, a two-part interview was carried out with an HCI expert involved in the human-led evaluation study of Rhino-Cyt. In the first part of the interview, the expert was asked to evaluate the 30 tasks with reference to three metrics (i.e. *Utility*, *Precision*, *Completeness*) assigning to each one a score ranging from 1 (low) to 5 (high). Specifically:

- (1) *Utility*: relevance of the task for the usability assessment of the system.
- (2) *Precision*: clarity and level of detail of the instructions for the user.
- (3) *Completeness*: extent to which the task mentions the necessary pieces of information for the analysis.

In the second part of the interview, the expert was asked to provide comments on the outputs of the LLMs in order to collect qualitative data about the validity of the tasks generated by AI models compared to those created manually and to understand whether they can provide a significant contribution to the test the design process. The complete set of results is available in the repository *LLMs4HCD* [2].

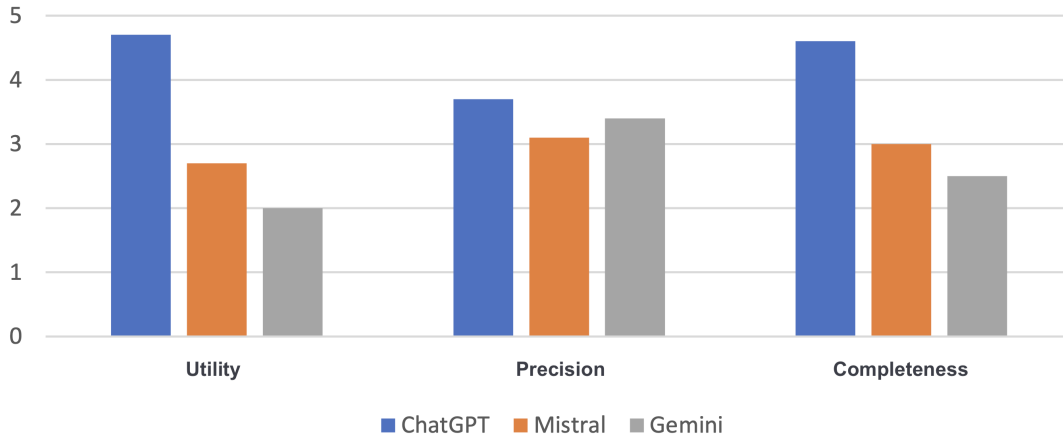


Figure 1: Scores for each tool based on the assessment of the interviewee

3 Results

The results presented in this section provide interesting quantitative and qualitative insights into the use of LLMs in the software development processes. Considering the temperatures and the models tested for each LLM the total number of tasks generated was: 300 for ChatGPT, 600 for Mistral, and 200 for Gemini. The full version of the table containing the tasks and their scores is available in the repository *LLMs4HCD* [2].

3.1 Task Feasibility

The model that generated the highest number of feasible tasks was GPT-4o with a temperature equal to 1, reaching 90%. Mistral's models peaked with mistral-medium-latest with 78% at a temperature of 0. Finally, Gemini's 1.5-pro model reached 81% of feasible tasks.

An exemplary set of tasks generated by GPT-4o is:

- (T1) View the results of the patient's last examination.
- (T2) View the list of cell images classified according to a category of your choice.
- (T3) Access the images of cells with a high probability of correctness and mark one of them as 'Incorrect'.
- (T4) Provide a correction to an incorrect cell classification, including the reasons for the correction.
- (T5) Review and validate the explanations provided by the system for the classification of a specific cell.

3.2 Interview with the Usability Expert

During the interview with the usability expert, the subset of 30 tasks containing the most similar ones was used (See Section 2.2). The expert assigned scores to each task for the three metrics to determine the overall reliability of the automatic generation of tasks through LLMs.

Figure 1 shows that the tasks of ChatGPT were consistently the best in terms of *Utility*, *Precision* and *Completeness* with the following scores: 4.7, 3.8, and 4.6. However, the tasks generated by Gemini were rated as least useful and least complete, with scores of 2 and 2.5, respectively. Although the Mistral models showed a notable performance, they were rated as the least precise with a

score of 3.1 in *Precision* [2]. ChatGPT appears to be the most reliable model for automatic task generation, with consistently high scores across all metrics. However, Mistral and Gemini can be performant depending on the considered metric.

After this first part of the interview, the HCI expert provided spontaneous feedback on the tasks he had evaluated. The researcher explained that these tasks would not be used as they were in a usability test, because some were too focused on a single aspect, others were too broad, and none addressed critical functionalities of the system.

The second part of the interview delved into the differences between the models. According to the interviewee, ChatGPT is the most useful model, followed by Mistral and Gemini. The interviewee pointed out that AI-generated tasks can be valuable in the brainstorming phase, but are not suitable as they are in the final stage of a usability test without final refinement. This underlines the need for human supervision to ensure that relevant scenarios are adequately addressed due to LLMs' potential biases and inability to understand properly the specific requirements that depend on the context.

4 Conclusions and Future Work

The findings of this study have highlighted both the potential and the limitations of LLMs in supporting experts to design usability tests. Among the models tested, ChatGPT consistently outperformed Mistral and Gemini in generating tasks that were consistent with the objectives of a usability test and system functionality. However, while LLMs can generate tasks for usability testing, the research underlines the importance of human oversight to refine these outputs and the correct prompts. Future work will include the use of the tasks generated by LLMs to conduct a usability study. It is also planned to investigate the integration of LLMs in all phases of the software development process to understand the extent to which humans can be assisted by this technology, fostering a human-AI symbiotic relationship. For generalizability purposes, the approach will be applied to different domains and systems.

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